

Spatial Data Lab

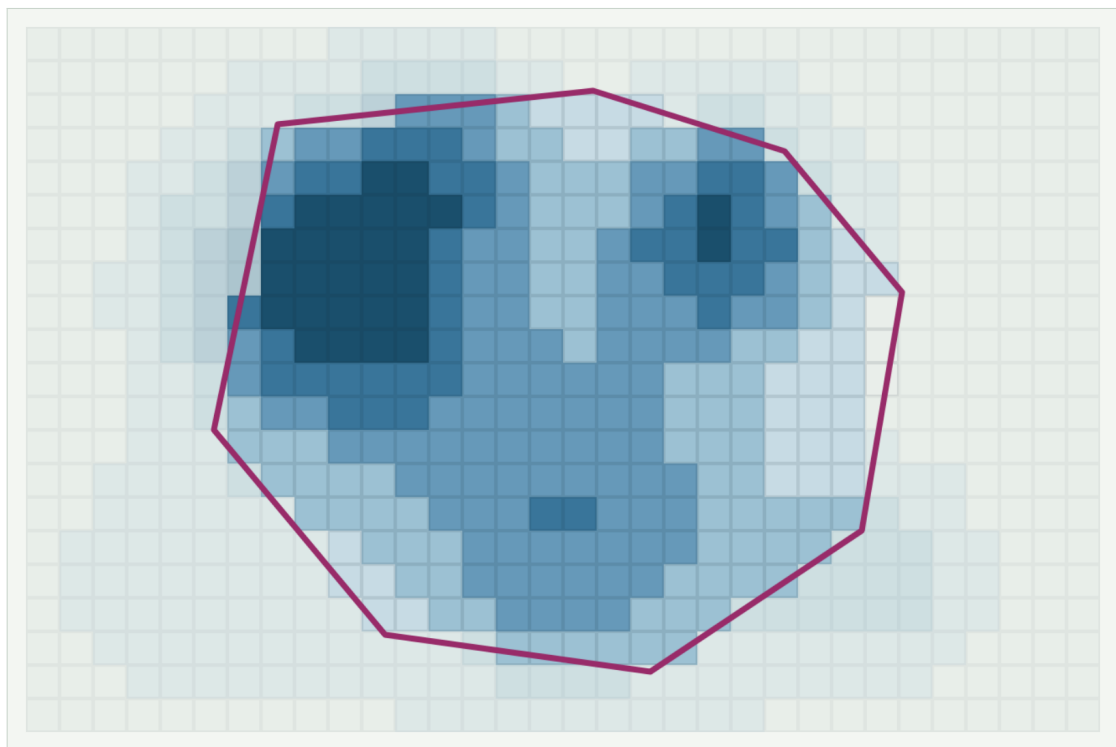
What the five modules show, and why each one matters

Spatial data goes wrong in five ordinary ways. None of them shows an error message. QGIS runs, a new column appears, and the number inside it is simply not what you think it is. The lab at billytulungen.com/lab lets you make each mistake on purpose and watch the figures change while you do. This handout explains what you are looking at in each one.

You do not need the lab open to read this, and you do not need to read this to use the lab. Together they cover the same ground twice, which is usually how something sticks.

The practice files are on the same page: twelve layers and tables, 430 KB, with a README naming which file each exercise needs.

1. How coarse a raster can get before the average stops meaning anything



What you are looking at

A satellite image of night lights. Every square is one cell, and every cell holds one number: how bright that patch of ground was. The magenta outline is a district.

QGIS can read every cell inside that outline and hand you a single figure for the district. The tool is called *zonal statistics*, and it is how a raster becomes a column in your dataset. Make

the cells larger and fewer of them fit inside the outline, so that single figure rests on less.

What to try

Drag the cell size to 16 km and count what is left inside the boundary. Compare the mean at 1 km with the mean at 16 km, then compare the standard errors beside them. Watch the sum while you drag: it swings hard, while the mean barely moves.

What it means

The mean is fairly robust to resolution; the sum is not, because a total depends on how many cells you happen to have. That is the practical argument for building rates rather than totals whenever you can.

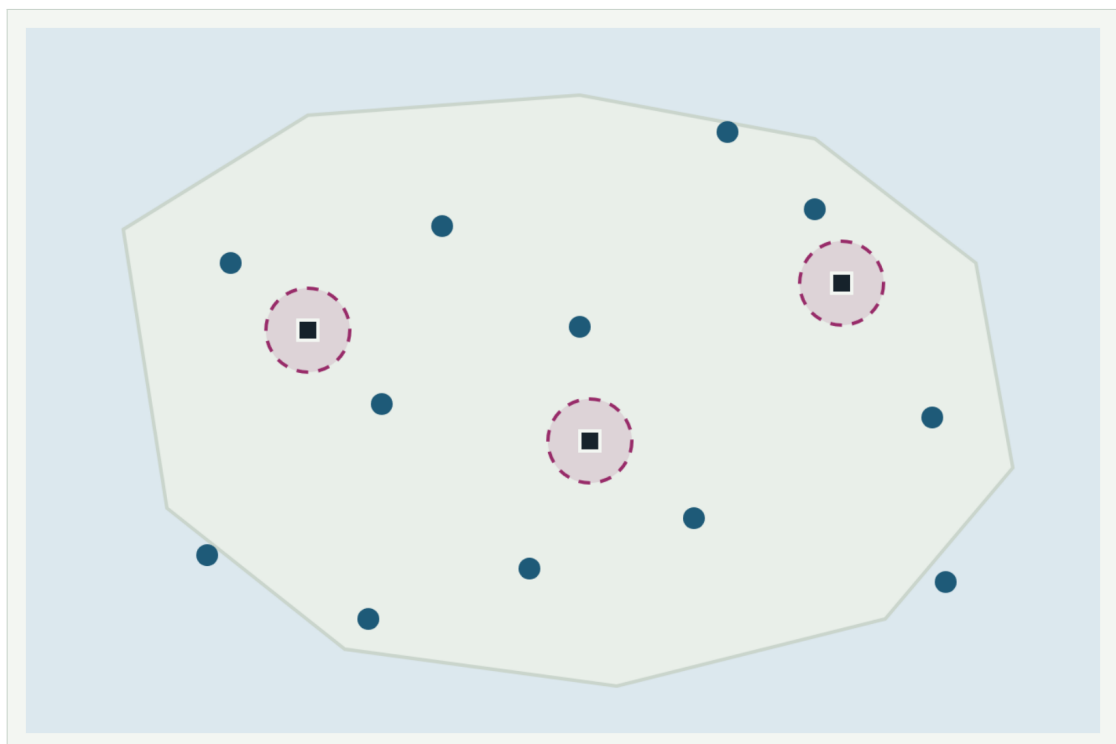
The standard error is the part most people skip. A district with twelve cells and a district with twelve hundred can report the same average, and nothing in the table tells the reader which is which.

In QGIS

Processing Toolbox > Zonal statistics

Tick `count` alongside `mean`. Count is how many cells went into each average. A small city district might contain only a dozen, and without that column nobody reading your results would ever know.

2. Why 5000 sometimes means five kilometres and sometimes five hundred



What you are looking at

Three villages, drawn as squares, and twelve health facilities, drawn as dots. You want to count the facilities within five kilometres of each village, so you draw a circle of that size around every village. In QGIS that circle is a *buffer*, and you set its size by typing a number.

Type 5000 and you might get five kilometres, or you might get roughly five hundred. Which one you get depends on the coordinate system the layer carries.

What to try

Leave the distance at 5 and switch the coordinate system to degrees. Read the radius you actually got. Then look for any value that still gives a sensible answer, because there isn't one.

What it means

A coordinate system stores position either in metres or in degrees of latitude and longitude. A degree is an angle, not a length. One degree of latitude covers about 111 km, so a buffer of 5 on a degree-based layer is a circle roughly 555 km across.

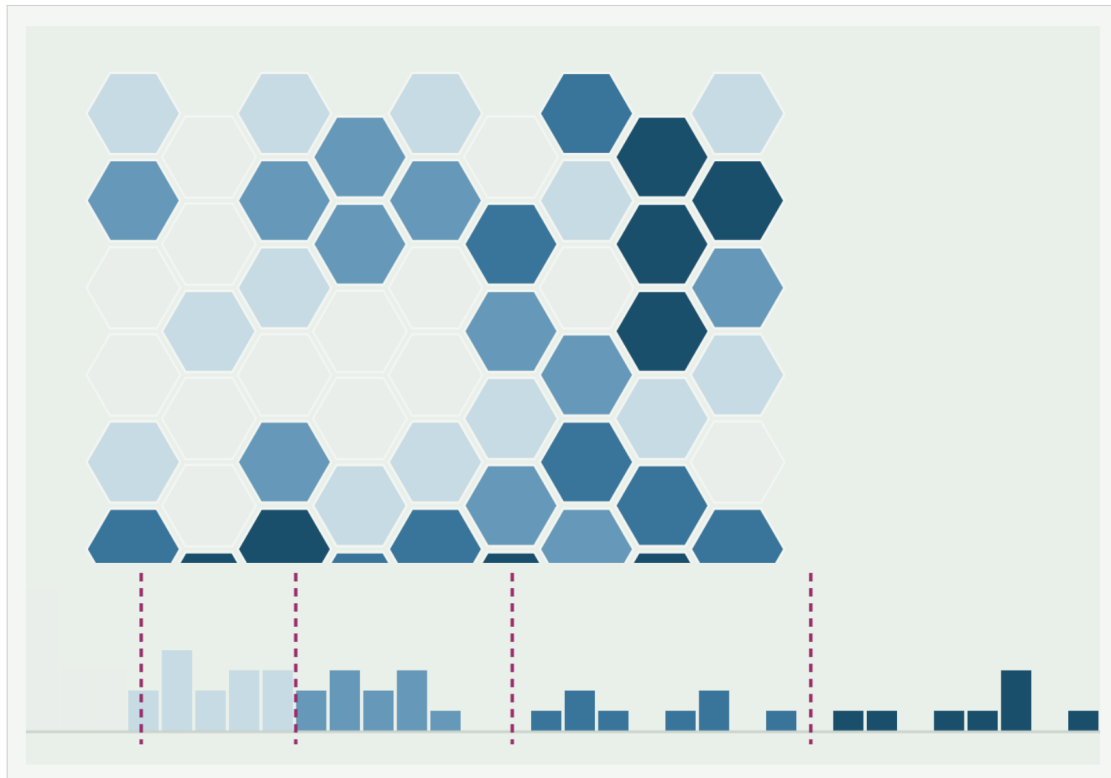
Worse, a degree of longitude gets shorter as you move away from the equator, so the error is not even the same size across your map. And once every village reaches every facility, your new column holds the same number everywhere. A column with no variation tells a regression nothing at all.

In QGIS

[Reproject Layer](#) > [Buffer](#) > [Join Attributes by Location](#)

Before anything that measures distance or area, reproject your data into the UTM zone covering your study area, which puts it in metres. Indonesia spans nine zones; Java and Kalimantan sit mostly in 48S and 49S, codes [EPSG:32748](#) and [EPSG:32749](#). Then check the EPSG code in the bottom-right corner of the QGIS window before you press Run.

3. One column of numbers, and how different the maps can look



What you are looking at

Fifty-four areas, each with one poverty figure, and a histogram underneath showing how those figures are spread. The dashed lines on the histogram are the class breaks: the points where one colour stops and the next begins.

Those figures never change. What changes is how you slice them. To colour a map you sort the values into groups, called *classes*, then give each class a shade. Where you cut between classes is your choice, not the data's, and so is the colour scheme.

What to try

Switch to Quantile and watch where the break lines land. Change the variable from the poverty rate to the number of poor people and ask whose map that actually is. Pick the rainbow scheme and try to rank the classes from low to high using colour alone.

What it means

Quantile forces roughly the same number of areas into every class, so the breaks bunch up wherever the data is dense. A map of nearly identical values ends up looking dramatic. Equal Interval splits the value range evenly instead, which on skewed data often leaves the top class holding one or two areas.

Mapping a count rather than a rate reproduces the population distribution whatever the title says. And a rainbow scheme has no natural order, so readers cannot tell which end is high, and readers with red-green colour blindness cannot read it at all.

In QGIS

[Layer Properties](#) > [Symbology](#) > [Graduated](#)

When a map goes into a paper, write the method and the number of classes into the caption. Without them a reader cannot tell whether the pattern is in the data or in your choices, and a referee will ask.

4. The join that drops part of your sample without saying so

What you are looking at

A *join* is how you attach a table of statistics to a map. Both sides need a shared key, usually the district code, and QGIS matches them row by row. The module shows twelve districts and twelve rows of statistics, and lets you switch on everyday situations that break the match.

The five ways it fails

Symptom	Cause	Fix
Every joined column is empty	The key is text on one side and a number on the other	Make the types match with the Field Calculator
Some rows are empty	Codes changed when districts split	Use a crosswalk table between years
Leading zeros disappear	Excel read the code column as a number	Import the CSV with the key column forced to text
Row count grows	The key is not unique on one side	Check for duplicates before joining
Names do not match	Spelling, abbreviation and capitalisation differ	Join on codes, never on names

What it means

The rows that fail are never a random handful. They are the newly created districts, the cities whose names changed, the areas that merged. Those places differ from the ones that matched, often in ways related to whatever you are studying.

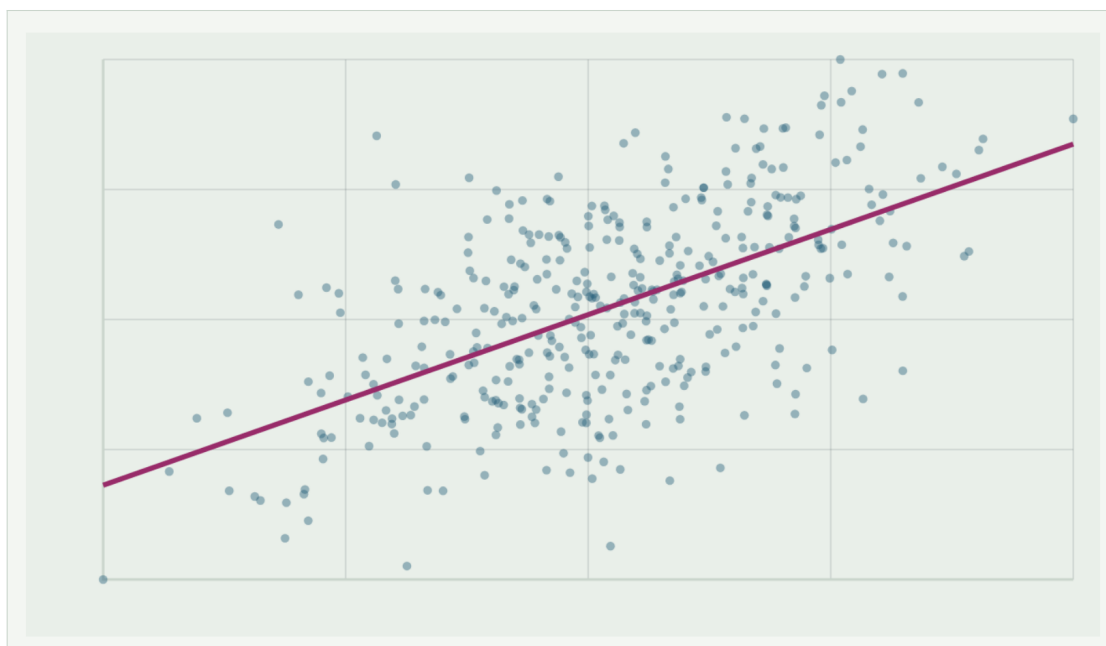
So this is not a tidying-up problem. It is a sample selection problem sitting quietly inside your results, and no error message will ever point at it.

In QGIS

[Layer Properties](#) > [Joins](#)

Count your rows before and after every join. Sort the new column to find the blanks. Match on codes rather than names. A join made this way is temporary until you run [Export](#) > [Save Features As](#), which freezes it into a permanent file.

5. What happens to a coefficient when you redraw the boundaries



What you are looking at

Four hundred households. The true relationship between road access and spending is fixed and never changes in this module.

Now group those households into areas and average each area, which is what you do whenever you work with district data instead of household data. Watch the slope of the line as the areas get larger. No new information arrives.

What to try

Note the coefficient at household level, then at four areas, and work out the multiple. At sixteen areas, switch the boundary shift on and off a few times: same households, different lines on the map, different answer. Watch R^2 climb as you aggregate.

What it means

Averaging cancels whatever is peculiar to each household but keeps whatever varies smoothly from place to place. What survives is the geographic part of the relationship, and here that part is much stronger than the household part. That is why the slope grows.

R^2 climbs with it, which is worth noticing on its own: a tighter fit is not a truer answer.

Two separate things are happening. Larger areas change the answer, and moving the boundaries changes it again even when the number of areas stays the same. Together they are known as the *modifiable areal unit problem*, and there is no general cure.

In QGIS

Dissolve > Join Attributes by Location > Export to CSV

Choose your unit because you can explain why that is the right one for the mechanism you

have in mind, then report the result at a second unit so readers can see how much it moved.

Which file for which exercise

Exercise	Files
1 Open a layer, check its CRS	kabupaten.gpkg
2 Join a statistics table	kabupaten_polos.gpkg and statistik_kabupaten.csv
3 Thematic map	kabupaten.gpkg
4 Point layer from a CSV	village_coordinates.csv
5 Buffer and spatial join	desa.gpkg and jalan.gpkg
6 Zonal statistics	nightlights.asc and kabupaten.gpkg
7 Print layout	whatever you made in exercise 3

Everything in the bundle is stored in [EPSG:4326](#), which is degrees. Reproject to [EPSG:32749](#) before you measure anything.

Where the real versions come from

What you need	Where to get it
Boundaries and statistics	BPS, statistical area maps, with codes matching the published tables
Official base maps	Badan Informasi Geospasial, tanahair.indonesia.go.id
Roads and facilities	OpenStreetMap, though coverage varies a lot between regions
Night lights	VIIRS, monthly and yearly
Rainfall	CHIRPS, daily since 1981
Forest cover	Global Forest Change, 30 m, yearly
Elevation and slope	SRTM

Every figure in this handout was produced by the lab itself, running on synthetic data generated so the patterns read clearly. None of it describes a real place, and nothing computed from the practice files should end up in a thesis. Swap in the real sources above before anything goes into a paper.

Billy Thandy Tulungen · guest lecture, 21 September 2026 · billytulungen.com/lab